

Practice Quiz: Agents (Computational Neuroscience)

Name: __ Date: __

Part A: Multiple Choice

1. In the cortical hierarchy, higher cortical areas: A) Process faster-changing, more concrete features B) Represent slower-changing, more abstract contexts and goals C) Only process motor information D) Are smaller than lower areas
2. The basal ganglia's primary role in Active Inference is: A) Sensory processing B) Policy (action) selection based on expected free energy evaluation C) Memory storage D) Language processing
3. Dopamine in the Active Inference framework signals: A) Pleasure and reward B) The precision of expected outcomes — confidence in policy selection C) Pain and punishment D) The speed of processing
4. The thalamus implements: A) Memory consolidation B) Precision gating — boosting or suppressing signals to control what reaches the cortex C) Language production D) Emotion regulation
5. Forward models in the cerebellum predict: A) The weather B) The sensory consequences of your own actions C) Other people's behavior D) Abstract mathematical relationships
6. Cerebellar damage typically causes: A) Memory loss B) Clumsy, uncoordinated movement due to broken forward models C) Blindness D) Emotional flatness
7. Acetylcholine's role in Active Inference is to: A) Modulate sensory precision — controlling attention to incoming data B) Control muscle contraction C) Regulate body temperature D) Produce neural oscillations

Part B: Short Answer

1. A patient with Parkinson's disease has difficulty initiating movements but can speak, see, and think normally. Using the Active Inference framework, explain which brain structure and neurotransmitter are most affected, and why this specifically disrupts action initiation.
2. Draw or describe the basic loop: cortex $\rightarrow$ basal ganglia $\rightarrow$ thalamus $\rightarrow$ cortex. Explain what information flows at each step and how this implements policy selection.
3. Compare the roles of the cerebellum and the basal ganglia in action. The cerebellum generates forward models; the basal ganglia select policies. Give a specific example where both are working together (e.g., reaching for a cup of coffee).